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Question Paper Code: 242GXEST203399

Reg No: Name:

VIMAL JYOTHI ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to APJ Abdul Kalam Technological University)

B.TECH DEGREE S2 (R/S) EXAMINATION APRIL 2026 (2024 SCHEME)

GXEST203: FOUNDATIONS OF COMPUTING: FROM HARDWARE ESSENTIALS TO WEB DESIGN

(CC,CS,CU,CN,AD,EC)

Max. Mark: 60 **Duration: 2 Hour 30 Minutes**

PART A
Answer All Questions

Q No.	Questions	CO	Marks
1.	What is the purpose of a MAC address in a NIC?	CO1	3
2.	What is firmware and where is it typically stored?	CO1	3
3.	Which Little Man Computer instructions would be classified as data movement instructions?	CO2	3
4.	What is the base (or radix) of Binary, Decimal, Octal and Hexadecimal number systems?	CO2	3
5.	What is the purpose of the 'ls' command in Linux? Mention any two options used with it.	CO3	3
6.	What is DHCP and why is it important in networking?	CO3	3
7.	What are the key elements of effective web content creation?	CO4	3
8.	What are the different ways to apply CSS to an HTML document?	CO4	3

PART B
Answer All Questions

Q No.	Questions	CO	Marks
9.	a) Explain the main functions performed by device management in an operating system. (5 Marks) b) Explain the role of I/O controllers in device management with examples. (4 Marks)	CO1	9
	OR		
10.	a) Describe the structure and working of an HDD. (5 Marks) b) Describe the working of optical drives and distinguish them with HDDs and SDDs. (4 Marks)	CO1	9
11.	a) If $(L)_M$ represents a number L in base–M number system, then identify the correct ascending order of the following numbers after converting each of them into the decimal (base-10) number system: (5 Marks) i. $(11011.11)_2$ ii. $(25.5)_8$ iii. $(16.6)_{16}$ b) A data acquisition system converts sensor readings from decimal form into different base-k representations before storage. During conversion, a trainee writes the remainders in the same order as obtained and gets an incorrect result. i. Identify the mistake made by the trainee. ii. Explain the correct procedure for decimal to base-k conversion. (4 Marks)	CO2	9
	OR		
12.	Explain CPU fetch execute cycle for LOAD instruction.	CO2	9

13.	a) Explain the DORA (Discovery, Offer, Request, Acknowledge) process in DHCP with an example. (5 Marks) b) What are three common DHCP issues and how can they be resolved? (4 Marks)	CO3	9
	OR		
14.	Describe the advantages and disadvantages of different network topologies with real-world applications.	CO3	9
15.	A company wants to improve the appearance of its webpage by adding a gray background, a centered blue heading and readable font styling across the page. a) Write suitable CSS rules to apply these visual styles. (5 Marks) b) Explain why using an external stylesheet is more efficient than inline styling for multi-page websites. (4 Marks)	CO4	9
	OR		
16.	a) Explain the role of FTP in transferring web files. How does FTP ensure efficient content management on a web server? (5 Marks) b) What are the necessary steps to upload files to a web server using FTP successfully? (4 Marks)	CO4	9

CO1: Identify the fundamental components and the working of an IT environment.
CO2: Explain the data representations, CPU architectures, and the basic functioning of a computer.
CO3: Explain the operating systems, computer network architecture, and necessary protocols used.
CO4: Develop simple interactive web pages and validate the inputs.

***Level:** Knowledge level based on Blooms Taxonomy
[L1.Remembering(R), L2.Understanding(U), L3.Applying(P)]